

Parking Management System

Overview

The **Car Parking Management System** is a console-based application developed in the C programming language to automate the management of vehicle parking within a parking facility. The system provides an efficient way to allocate parking slots, maintain vehicle records, calculate parking charges based on parking duration, and generate parking receipts.

Instead of maintaining parking information manually, the application stores all vehicle details electronically and allows users to perform parking operations through a menu-driven interface. The system also maintains parking records using binary file handling, ensuring that vehicle information is preserved even after the application is closed.

The application follows a modular programming approach, where different functionalities such as parking operations, queue management, and file handling are implemented in separate source files. This improves readability, maintainability, and scalability of the project.

The project is designed to simulate a real-world parking management environment while demonstrating the practical application of core C programming concepts such as structures, arrays, functions, file handling, queues, enumerations, pointers, and modular programming.

Key Features

- Automatic parking slot allocation using Queue data structure.
- Vehicle entry and exit management.
- Parking charge calculation based on vehicle type and parking duration.
- Automatic Parking ID generation.
- Vehicle search using registration number.
- Parking receipt generation.
- Parking status monitoring.
- Display of available parking slots.
- Persistent data storage using binary files.
- Admin authentication for revenue report access.
- Menu-driven user interface.

Scope of the Project

The project is intended for small and medium-sized parking facilities where vehicle records need to be managed efficiently. It reduces manual effort, minimizes human error, and provides quick access to parking information.

The current version supports:

- Vehicle Registration

- Slot Allocation
- Vehicle Exit
- Parking Charge Calculation
- Record Searching
- File Storage
- Revenue Management (Admin Access)

Future enhancements may include online booking, barcode/QR code integration, database connectivity, and graphical user interface support.

Project Objective

To develop a **Console-Based Car Parking Management System** in C that automates vehicle parking operations, efficiently manages parking slots, calculates parking charges, and maintains parking records using file handling.

Specific Objectives

- ✓ Automate parking slot allocation.
 - ✓ Reduce manual record keeping.
 - ✓ Store vehicle information securely using file handling.
 - ✓ Calculate parking charges based on vehicle type and parking duration.
 - ✓ Provide quick vehicle search using vehicle number.
 - ✓ Maintain accurate parking history.
 - ✓ Display current parking status and available slots.
 - ✓ Generate parking receipts for customers.
 - ✓ Protect sensitive financial information through admin authentication.
 - ✓ Improve operational efficiency and reduce parking management errors.
-

Goals

Goal	Description
Efficient Parking	Allocate available parking slots automatically.
Record Management	Maintain complete vehicle records.
Data Persistence	Save records permanently using binary files.
Accurate Billing	Calculate parking charges automatically.
Security	Restrict revenue information to authorized admin only.
User Convenience	Simple menu-driven interface for easy operation.

Expected Outcomes

After successful implementation, the system should be able to:

- Allocate parking slots automatically.
 - Manage vehicle entry and exit.
 - Search parked vehicles quickly.
 - Display current parking occupancy.
 - Calculate parking fees accurately.
 - Generate parking tickets and receipts.
 - Save and retrieve parking records.
 - Display available parking slots.
 - Protect admin features using password authentication.
-

Advantages of the Proposed System

- Faster than manual parking management.
 - Reduces human errors.
 - Easy to operate.
 - Organized record keeping.
 - Automatic billing.
 - Efficient parking space utilization.
 - Secure access to revenue reports.
 - Scalable for future enhancements.
-

Objective Summary

Objective	Status
Automatic Slot Allocation	Achieved
Vehicle Registration	Achieved
Parking Charge Calculation	Achieved
Vehicle Search	Achieved
Parking Status Monitoring	Achieved
File Storage	Achieved
Receipt Generation	Achieved
Admin Authentication	Achieved

User Stories

User stories describe the system requirements from the perspective of different users. They help define what the system should accomplish and the expected outcome for each operation.

User ID	User Story	Acceptance Criteria	Priority
US-01	As a parking operator, I want to park a vehicle so that an available parking slot is automatically assigned.	The system should allocate the next available parking slot and generate a unique Parking ID.	High
US-02	As a parking operator, I want to record vehicle details so that every parked vehicle is stored in the system.	Vehicle number, owner name, vehicle type, entry time, and slot number should be stored successfully.	High
US-03	As a parking operator, I want to prevent duplicate vehicle entries so that the same vehicle cannot occupy multiple parking slots.	The system should detect duplicate vehicle numbers and reject duplicate entries.	High
US-04	As a parking operator, I want to search for a vehicle using its registration number so that I can quickly locate it.	The system should display complete vehicle details if the vehicle exists; otherwise, an appropriate message should be displayed.	High
US-05	As a parking operator, I want to process vehicle exit so that parking charges can be calculated automatically.	The system should calculate the parking duration, generate the bill, free the parking slot, and update records.	High
US-06	As a parking operator, I want to view all parked vehicles so that I can monitor the current parking occupancy.	The system should display a list of all vehicles currently parked.	Medium
US-07	As a parking operator, I want to view available parking slots so that I know how many slots are currently free.	The system should display all available slot numbers along with the total number of free slots.	Medium
US-08	As a parking operator, I want the parking records to be saved automatically so that	All parking records should be saved to a file and	High

	data is not lost after the program closes.	restored when the application starts again.	
US-09	As an administrator, I want to access the revenue report using a password so that financial information remains secure.	Only the correct password should allow access to the revenue report.	High
US-10	As an administrator, I want to view the total revenue and number of vehicles served so that I can monitor the parking business performance.	The system should display accurate revenue statistics after successful authentication.	Medium

Priority Classification

Priority	Description
High	Essential functionality required for the system to operate correctly.
Medium	Important features that improve usability and management but are not critical for basic operation.
Low	Optional enhancements that can be added in future versions.

Functional Requirements

FR-01: Vehicle Parking

- Accept vehicle number, owner name, and vehicle type.
 - Allocate the next available parking slot.
 - Generate a unique Parking ID.
 - Record the vehicle entry time.
-

FR-02: Parking Slot Allocation

- Allocate slots using the queue (FIFO).
 - Assign only available slots.
 - Prevent allocation when parking is full.
 - Update available slot count.
-

FR-03: Vehicle Exit

- Search the parked vehicle.
 - Record the exit time.
 - Calculate parking charges.
 - Release the occupied parking slot.
-

FR-04: Vehicle Search

- Search using vehicle number.
 - Support case-insensitive search.
 - Display complete vehicle details.
 - Show an error message if not found.
-

FR-05: Display Parked Vehicles

- Display all currently parked vehicles.
 - Show Parking ID, vehicle number, owner name, and slot number.
 - Display entry time and vehicle type.
-

FR-06: Parking Status

- Display total parking capacity.
 - Display occupied slots.
 - Display available slots.
 - Display current occupancy status.
-

FR-07: Available Slot Display

- Display all free parking slot numbers.
 - Display total available slots.
 - Update information after every entry and exit.
-

FR-08: Parking Charge Calculation

- Calculate parking duration.
 - Apply charges according to vehicle type.
 - Charge a minimum of one hour.
 - Display the total parking fee.
-

FR-09: File Management

- Save parking records to a binary file.
 - Load records when the program starts.
 - Preserve data after program termination.
-

FR-10: Revenue Management

- Maintain total revenue collected.
 - Maintain total vehicles served.
 - Display revenue statistics.
-

FR-11: Admin Authentication

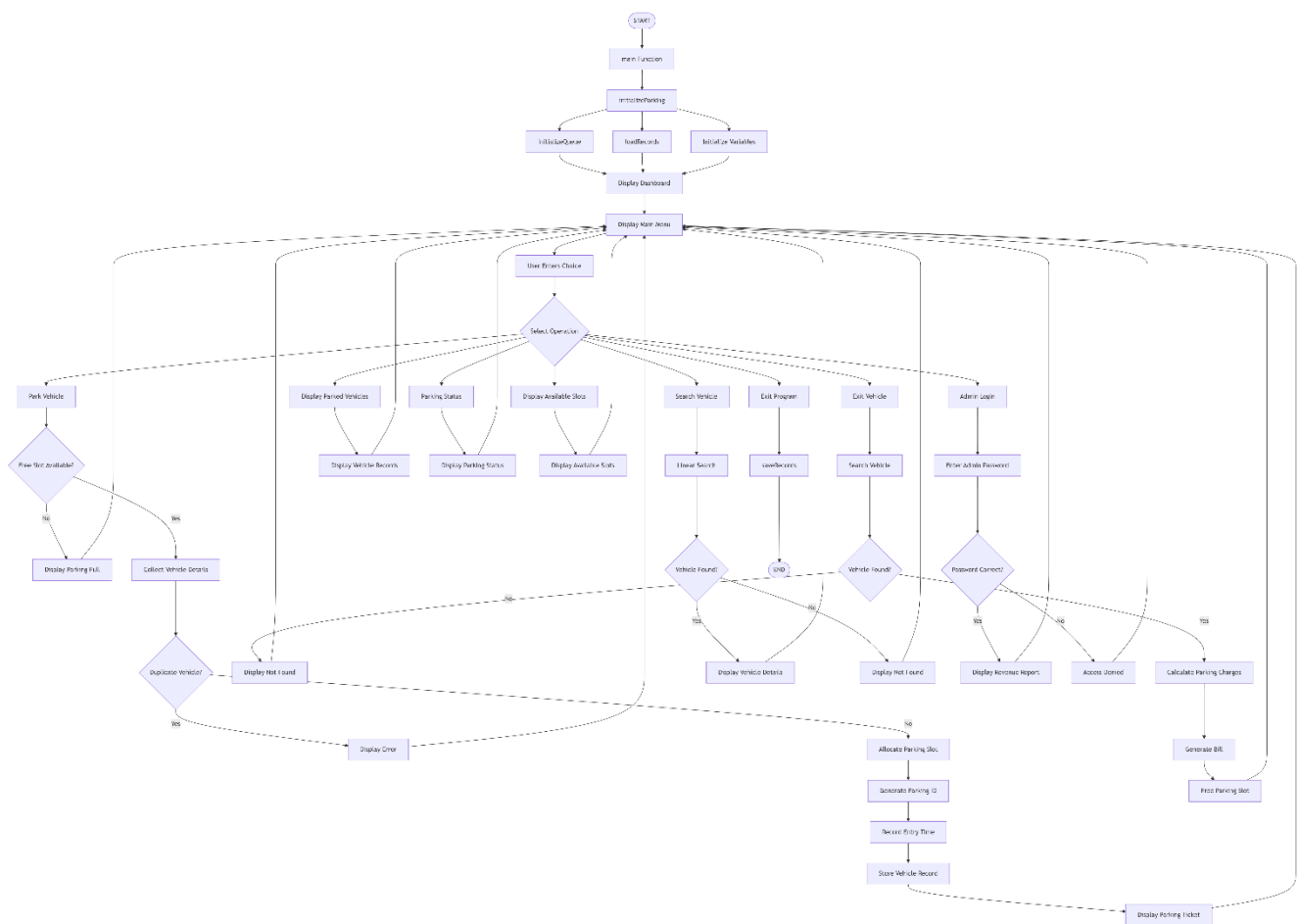
- Request administrator password.

- Validate the entered password.
- Allow access only after successful authentication.
- Restrict revenue report from general users.

Non-Functional Requirement

Requirement ID	Category	Description
NFR-01	Performance	Fast response and efficient processing
NFR-02	Reliability	Accurate and consistent system operations
NFR-03	Usability	User-friendly console interface
NFR-04	Data Integrity	Accurate and valid parking records
NFR-05	Security	Password-protected admin features
NFR-06	Maintainability	Modular and well-organized code
NFR-07	Scalability	Easy expansion of system features
NFR-08	Portability	Compatible with standard C compilers
NFR-09	Availability	Ready for use with automatic record loading
NFR-10	Data Persistence	Permanent storage using file handling

Parking Management System - Code Flow



Concepts Used

The project implements various C programming concepts and software development practices.

Concept	Application in Project
Structures (struct)	Used to store vehicle and parking information
Arrays of Structures	Used to maintain multiple vehicle records
Functions	Provides separation of different operations
Modular Programming	Divides the project into multiple source files
File Handling	Used for permanent record storage
Date and Time (time.h)	Records vehicle entry and exit time
Searching	Used for locating vehicle records
Queue Data Structure	Used for automatic slot allocation

Input Validation	Prevents invalid user entries
Menu-Driven Design	Provides simple user interaction
Enumeration	Used for vehicle type representation
Pointers	Used for passing structures and queues to functions

Main Functions

The system is implemented using separate functions with clear responsibilities.

Function	Purpose
initializeParking()	Initializes parking system variables and loads records
initializeQueue()	Creates and manages available parking slots
parkVehicle()	Registers a vehicle and assigns a slot
allocateSlot()	Provides the next available parking slot
searchVehicle()	Searches vehicle records using vehicle number
exitVehicle()	Processes vehicle exit and billing
calculateCharge()	Calculates parking fee based on duration and vehicle type
displayParkingStatus()	Shows current parking occupancy
displayAvailableSlots()	Displays free parking slots
displayRevenueReport()	Shows revenue details for admin
saveRecords()	Stores records permanently
loadRecords()	Retrieves saved records
generateParkingID()	Creates unique Parking IDs

Data Structures Used

Vehicle Structure

The Vehicle structure is used to store complete information about a parked vehicle.

```
struct Vehicle
{
    int parkingID;
    char vehicleNumber[20];
    char ownerName[50];
    int type;
    int slotNumber;
    time_t entryTime;
```

```
time_t exitTime;  
double parkingCharge;  
int isParked;  
};
```

Stored Information:

- Unique Parking ID
 - Vehicle registration number
 - Owner name
 - Vehicle type
 - Allocated parking slot
 - Entry time
 - Exit time
 - Parking charge
 - Parking status
-

Slot Management Structure

The parking slot information is managed using a queue-based approach.

```
struct Slot  
{  
    int slotNumber;  
    int availability;  
};
```

Purpose:

- Maintain parking slot information.
- Track available and occupied slots.
- Support automatic slot allocation.

Expected Program Features

The Parking Management System is designed to provide an efficient and user-friendly solution for managing vehicle parking operations. The system automates parking slot allocation, maintains vehicle records, calculates parking charges based on parking duration, and generates reports for effective parking lot management. The following table describes the expected features of the proposed system.

Feature	Description
User-Friendly Menu	Provides a simple command-line interface (CLI) with a menu-driven system for easy navigation between different operations.
Automatic Slot Allocation	Automatically assigns the first available parking slot to a vehicle using a Circular Queue, eliminating manual slot selection.
Vehicle Registration	Records essential vehicle details such as Parking ID, Vehicle Number, Owner Name, Vehicle Type, Entry Time, and Allocated Slot Number.
Unique Parking ID Generation	Generates a unique Parking ID automatically for every newly parked vehicle to ensure easy identification.
Vehicle Search	Allows searching for parked vehicles using the vehicle registration number through Linear Search.
Duplicate Vehicle Detection	Prevents the same vehicle from being parked multiple times by checking existing parking records.
Parking Slot Management	Continuously maintains the status of available and occupied parking slots and updates them after every vehicle entry and exit.
Vehicle Exit Management	Records the exit time of the vehicle, calculates parking duration, generates the parking bill, and releases the occupied slot for future use.
Automatic Parking Charge Calculation	Calculates parking charges based on the vehicle type and total parking duration using predefined hourly rates.
Parking Receipt Generation	Generates a detailed parking receipt containing Receipt Number, Parking ID, Vehicle Details, Entry Time, Exit Time, Slot Number, and Total Charges.
Parking Status Display	Displays the total parking capacity, occupied slots, available slots, and parking occupancy percentage.
Current Parked Vehicle List	Displays a formatted table containing details of all vehicles currently parked in the parking lot.

Revenue Management	Maintains the total revenue collected from parking charges and the total number of vehicles served.
Data Validation	Validates user inputs such as vehicle number and vehicle type to reduce incorrect data entry.
Error Handling	Displays appropriate error messages for invalid operations such as parking when full or attempting to remove a vehicle that is not parked.
Data Storage	Stores parking records in a data file (parking.dat) using file handling so that records can be preserved for future use.
Modular Programming	Implements the project using multiple source and header files, making the program easier to understand, maintain, and extend.

Test Cases

The following test cases were performed to verify the correctness and functionality of the Parking Management System.

Test Case ID	Scenario	Input	Expected Output	Status
TC-01	Park a new vehicle	Vehicle No: DL01AB1234, Owner: Rahul, Type: Car	Parking ID generated, slot allocated, parking ticket displayed	Pass
TC-02	Park another vehicle	Vehicle No: UP32XY5678, Owner: Aman, Type: Bike	Next available slot allocated successfully	Pass
TC-03	Attempt to park the same vehicle again	Vehicle No: DL01AB1234	Message displayed: "Vehicle Already Parked"	Pass
TC-04	Parking when all slots are occupied	Valid vehicle details	Message displayed: "Parking is Full"	Pass
TC-05	Search for an existing vehicle	Vehicle No: DL01AB1234	Complete vehicle details displayed	Pass
TC-06	Search for a non-existing vehicle	Vehicle No: HR26AA9999	Message displayed: "Vehicle Not Found"	Pass
TC-07	Exit a parked vehicle	Vehicle No: DL01AB1234	Parking bill generated, slot released, revenue updated	Pass
TC-08	Exit a vehicle that is not parked	Vehicle No: HR26AA9999	Message displayed: "Vehicle Not Found"	Pass
TC-09	Verify parking charge calculation	Vehicle parked for 2 hours (Car)	Parking charge calculated as ₹100	Pass
TC-10	Display current parked vehicles	Select "Display Parked Vehicles"	List of all currently parked vehicles displayed in tabular format	Pass

TC-11	Display parking status	Select "Parking Status"	Total slots, occupied slots, available slots, and occupancy percentage displayed	Pass
TC-12	Display revenue report	Select "Revenue Report"	Total vehicles served and total revenue displayed	Pass
TC-13	Enter invalid vehicle type	Vehicle Type: 7	Error message: "Invalid Vehicle Type"	Pass
TC-14	Enter invalid vehicle number	Vehicle Number: A1	Error message: "Invalid Vehicle Number"	Pass
TC-15	Check slot reallocation	Exit vehicle from Slot 5, then park a new vehicle	Freed slot becomes available and is allocated correctly through the queue	Pass
TC-16	Save parking records	Exit program after operations	Vehicle records successfully saved to parking.dat	Pass
TC-17	Load existing records	Restart the application	Previously saved parking records loaded successfully	Pass
TC-18	Display dashboard	Start the application	Dashboard displays total slots, occupied slots, available slots, revenue, and vehicles served	Pass